

ENERGY AUDITS AND PROJECTS DO MORE THAN SAVE ENERGY

Brian Doucette P Eng
Manager of Energy Optimization
Suncor Energy Company
Fort McMurray, Alberta, Canada

James Robinson, P Eng, CEM
Principal Project Engineer
Automation Applications Inc, LLC
Exton, Pennsylvania, USA.

ABSTRACT

As a result of an Energy, Process and Control System Audit, Suncor Energy began an Advanced Control System Project at the Oil Sands Extraction Unit in Alberta, Canada. The methods and initial results of that ongoing project are detailed in this paper. The initial step was to conduct an audit of the existing process units and their related controls. The audit's objective was to identify ways to reduce energy consumption, reduce green house gas (GHG) emissions, improve product quality and increase production. Several teams familiar with the process areas worked together to develop Control Objectives Analysis (COA) and related design packages utilizing improved controls, production modifications and modified operating procedures to meet the rapidly changing demands placed on energy producers. The Advanced Process Control System Project consists of two distinct parts. At the regulatory level teams reviewed and improved existing systems. The advanced control group developed a process model that was an effective method to identify process opportunities, obtain input from other team members and validate the resulting process modifications and control strategies prior to implementation. As a result of the combined efforts of small high performance groups of Suncor employees, system integrators and equipment vendors we anticipate a net reduction in energy usage, net reduction in GHG emissions, improved recovery yields and increased production generating a net positive cash flow impact of C\$40MM/year.

INTRODUCTION

Suncor Energy is the first successful oil sands extraction facility in the world. Oil sands are unconsolidated oil sand deposits impregnated with petroleum, normally referred to as bitumen. Alberta has the world's largest deposit of oil sands estimated at 1.3 trillion barrels¹. The

Suncor facility uses several processes to mine ore and produce high quality synthetic crude oil for refining at other facilities. This process is described in the following section. The project discussed here focused on the Extraction Business Unit. Extraction is responsible for the separation of bitumen from the ore. It is also the largest user of thermal and electric energy at the facility. An energy process and control system audit began in September of 2000. An advanced control project started in the second quarter of 2001.

The topic presented here is that the mechanism of executing energy audits and advanced control system projects can result in more than energy cost reduction. An example is the DMI Paper Mill in Peace River, Alberta Canada where the application of an energy control system reduced the electric and gas purchases by 13% to 21% while stabilizing the steam header to the process units² improving product quality. A similar situation is presented on this project where the focus was stabilizing the hydrotransport feed rate, which resulted in a C\$8.7MM/year savings in hot water cost.

PROCESS DESCRIPTION

Suncor Energy continues to develop successful methods to economically extract bitumen from the oil sands. Currently, the near surface bitumen deposits allow Suncor to use mining techniques. A simplified process diagram is shown in Figure 2 – Process Diagram.

Mining

The mining process has evolved from the original dragline, bucketwheel and conveyor mining to a more energy efficient shovel, truck and hydrotransport method. This ultramodern mining complex uses several classes of shovels and a fleet of heavy hauler trucks to extract the ore from the mine face and deliver it to the sizers at Ore Processing Plants (OPP). Loaded heavy

haulers weigh over one million pounds and the magnitude of equipment is shown in Figure 1. The shovels and trucks are equipped with computers, GPS, and telecommunication equipment to communicate with mine dispatch. The Mine Dispatcher's job resembles that of an air traffic controller as he orchestrates all the shovels and trucks to assure delivery of the proper tonnage and ore quality to the OPP's.



Figure 1 - Heavy Hauler

Ore Processing Plant (OPP)

The extraction process is a version of the Clark Hot Water Extraction (CHWE)³ process

modified to operate at lower temperatures and with a minimum of chemical additives. Ore is received and crushed at the sizer at the OPP and conveyed to the rotary breaker where it is mixed with hot water, collected in the slurry storage tanks and pumped by three 2500HP pumps in series through the hydrotransport lines to the extraction plant.

Extraction

Extraction receives the ore from the hydrotransport and separates the bitumen from the sand called tailings (sand and other residue). The bitumen then travels through the dearator, interstage tank, froth treatment plant and to the diluted bitumen storage tank. The tailings travel to the consolidated tailings system to facilitate water removal and is pumped to the tailings ponds to settle out the remaining solids.

Upgrading

The final process unit is the upgrader that converts bitumen to a high quality synthetic crude oil. The synthetic crude is transported to refineries to the South for processing into petroleum products.

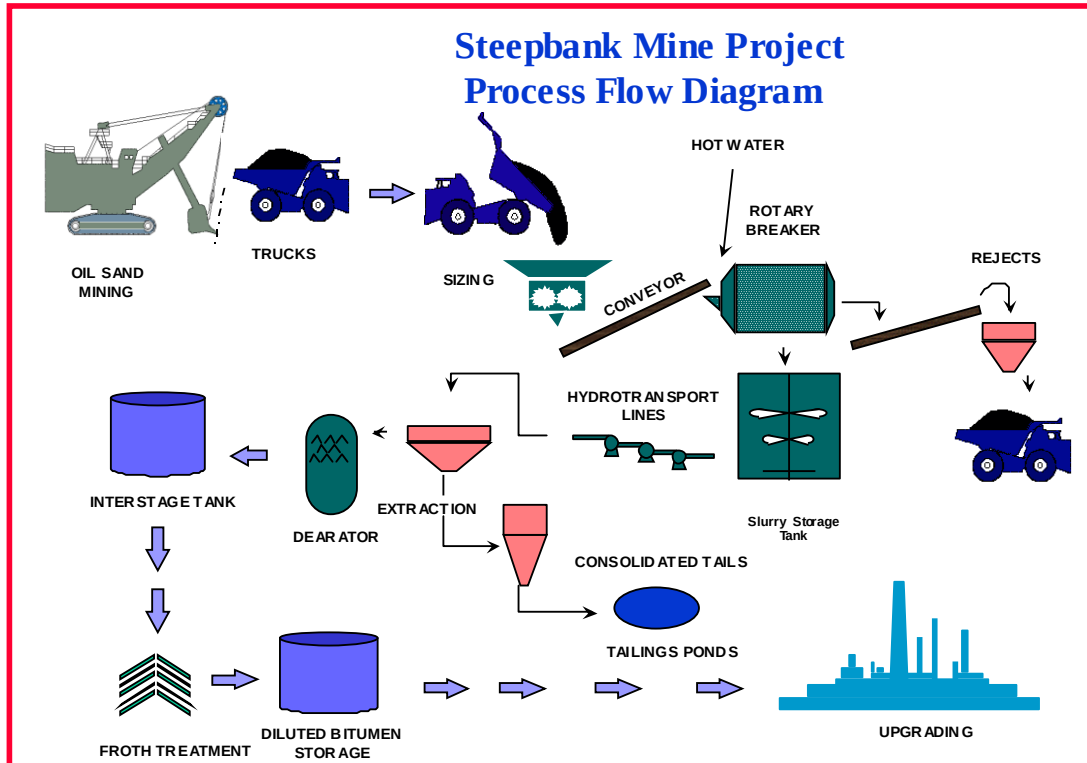


Figure 2 - Process Diagram

METHODOLOGY

Suncor is undergoing an intensive capital expansion program that necessitated a review of the existing extraction process unit. The initial requirements were an Energy, Process and Control System Audit to identify opportunities for process improvements. Upon completion of the audit, an Advanced Control System Project was initiated.

Audit

Process and control system audits take many different forms based on the client's requirements. The methodology used in this audit was detailed in an earlier paper⁴ and targets the needs defined by the client. The traditional approach for an effective audit contains the following steps:

- Project Definition: definition of client objectives
- "As-Is" System Status: clearly define the system as it stands.
- Analysis: defined by the client objectives, develop Control Initiatives with savings
- Verify Findings: validate the findings and verify the results with the client.
- Issue a Report: finalize the system analysis based on project objectives and verification issues.

But this audit was unique in two ways. First, Suncor was undergoing an intensive capital expansion and debottlenecking program to increase production capacity and product quality. The aggressive nature of the project execution resulted in process and operating changes overnight that would negate previous or proposed audit work. Second, the employees at Suncor are aggressive and empowered. There is constant drive at all levels to identify and improve operating conditions. To respond to these dynamics, the audit team developed informal high performance work groups on an area-by-area basis. The work groups consisted of representatives from operations, maintenance, project administration, process engineering, instrument and electrical technicians, control systems engineering, information technologies and the training department. This free exchange of information allowed the audit team to be more effective and focus on areas of new work while supporting existing work. This approach

allowed for the development of a more accurate work product:

- Control Initiatives were used to document the different projects that were discussed. The depth of knowledge from the work groups improved the accuracy of the Control Initiatives.
- Data Sources were expanded to include the Equipment Loss, Process Loss and Incident Reporting System in order to capture the history and cost of past incidents
- A program called Total Plant© was used to provide an analysis of the process control status and determine the effects of process upsets.

Project Execution

A review of the project initiatives indicated that improvements in production and density control through the hydrotransport system would result in C\$40MM/year which could justify the balance of the energy and cost avoidance initiatives. An Advanced Control System Project was approved to implement the primary initiatives and execute the secondary initiatives recommended by the work groups. The teams working on the audit were moved to the project for the following reasons:

- To work the best resources familiar with the project and decrease project cycle time.
- To coordinate with ongoing multiple projects – more bang for the buck.
- To insure compliance with internal corporate policies.
- To promote cost sharing between teams and overlay system enhancements on projects already underway.

Due to the size and complexity of the project, it was broken into two phases. Phase I (Figure 4) includes Train#2 starting at the mine and working into extraction. This allowed integrating the control modifications into a single line during process outages. This approach allowed validation of the modifications before integration into the other systems. Phase II will include integrating the new systems into Train#1 at the mine and the remaining portion of the extraction plant

CONTROL STRUCTURE

The Suncor Energy Oil Sands plant is aggressively integrating all aspects of the Mine and Extraction Plant Operation into a cohesive process management and reporting system. The primary objective of this project was to improve the controls in the base extraction plant. The base plant control system architecture is detailed in Figure 3 and provides a secure process control system while presenting strategic operating information to the business management and reporting systems. The control system hierarchy levels are summarized below:

- The Process Connected equipment are devices such as transmitters, valves, variable speed drives, etc. that are used to measure and manipulate the process.
- The Distributed Control System (DCS) / Programmable Logic Control (PLC) provide the intelligence of the system to the Process Connected Equipment to

read the process variable inputs and manipulate the process output devices.

- Several Control Rooms house the Man Machine Interface (MMI) equipment necessary to view and manipulate these systems. These control centers are located throughout the complex for the use of front line operation and supervisory personnel.

Recently, a series of Advanced Process Management Initiatives such as the Production Rate Control (PRC)© and constraint reporting systems were initiated to support the business objectives of the Oil Sands facility.

Plant wide scheduling and reporting systems are supported through the Information Technologies (IT) groups. These systems bridge the process control and information technology domains using firewalls and other security techniques to prevent unauthorized manipulation of the process variables.

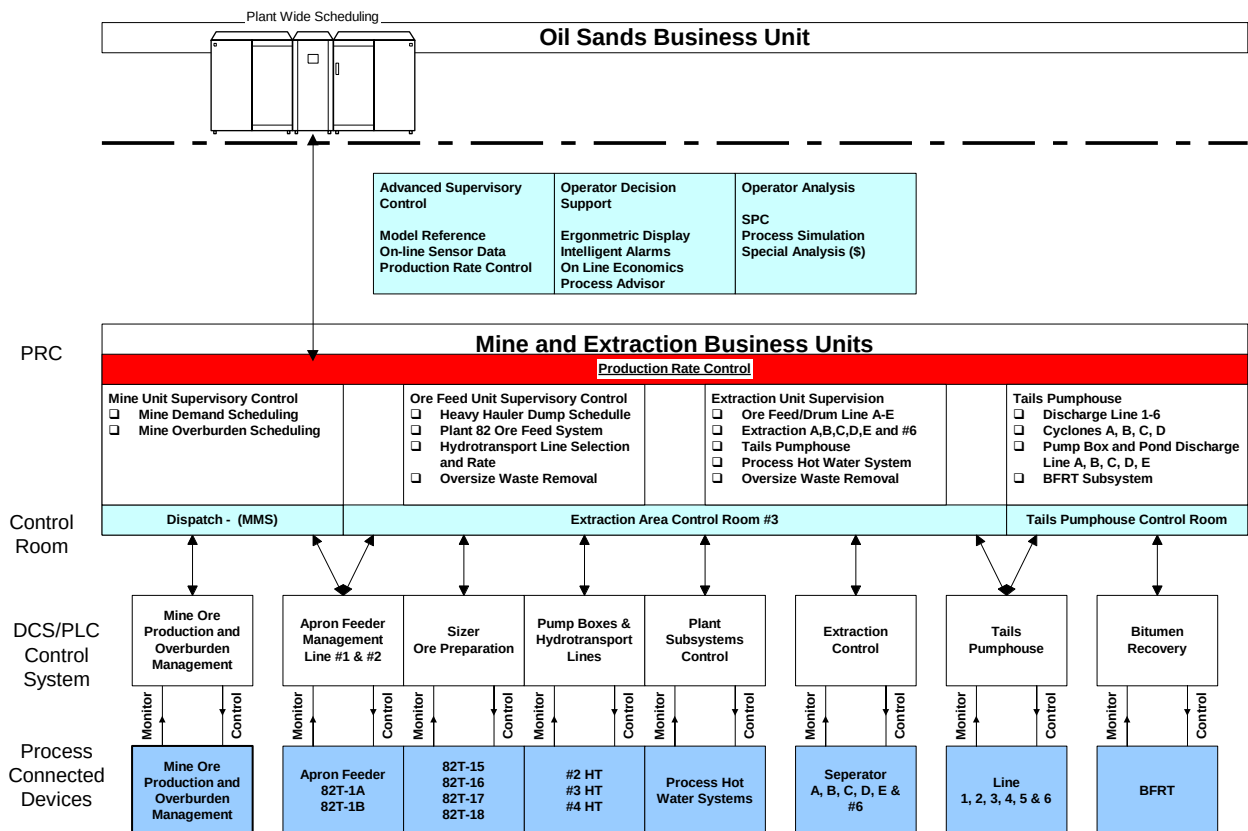


Figure 3 - Extraction Control System Architecture

Project Objectives

The audit indicated that substantial savings and production increases could be enjoyed through process modifications and control system enhancements. Due to the size and complexity of the project the effort was broken into two broad categories, Foundation Work and Advanced Control System. The overall structure of the project is presented in Figure 3 and detailed below:

Foundation Work

The primary objective of the Foundation Work was to document the existing process and control systems and work with the various teams to correct control problems. This became a formidable task due to the lack of up to date information. The following tasks were performed by the project teams:

- Documentation validation was mandatory. The area teams walked down and documented the process and instrument diagrams and provided process control narratives of the programs and logics. This included updating process flow diagrams, flowcharting process control programs and the development of SAMA diagrams of the existing control strategies. The format is designed to be compatible with the effort by a separate team that was developing training manuals.
- System Loading Study and Diagnostics (Figure 3 – Extraction Control System Architecture) were performed and a system loading study developed to assure overall system reliability.
- An overall system software cleanup was performed to remove inactive points and control programs.
- Inoperative control loops associated with the Advanced Control Platform were fixed. Once the overall system is operational, a systematic control loop evaluation and tuning will be accomplished.
- Several new technologies were integrated into the process to provide operations with the information they need to improve process controllability.
- Once the documentation was developed and organized, a mechanism was

developed to allow the DCS system to automatically update the system documentation. This assures the most current documentation to engineering, operations and the E&I groups.

Advanced Control System Work

As the foundation control system was documented and corrected, the implementation of advanced regulatory and supervisory control systems began. The specific objective was to apply the very best technology to the process. Significant work had been done by previous teams to get the plant on line and solve a variety of control problems. The current teams work towards a seamless integration of the existing and new applications to minimize operator retraining. A series of new technologies were developed to support Suncor's applications such as:

- Mine Dispatch/Distributed Control System interface to provide predictive tonnage available and anticipate ore quality to extraction. It will also return available production rates and process status from extraction to mine dispatch. The objective of the system is to match the performance of both areas and always push the available production capacity.
- Integration of the overall system into a Production Rate Controller (PRC)© to maximize the production rate. This allows extraction to move the process unit to match the available ore processing rate of the mine delivery system.
- Integration of new measurement and control technologies that provide more accurate and reliable control. This included parallel implementation of control initiatives by several separate project teams.
- A constraint reporting system responding online and in real time to help operations make better decisions.
- Enterprise management and scheduling to maximize equipment utilization.
- System diagnostics and reporting to allow proactive rather than reactive operations to abnormal operating conditions.

being automated.

CONTROL ALGORITHM

The overall Production Rate Control (PRC)© is used in the pulp and paper industry and is now being applied to the extraction process. The PRC© is detailed in Figure 4 below. The objective of this control strategy is to combine the best control technologies to meet the needs of the business unit. As the project developed, regulatory control strategies were integrated into the blocks marked “Regulatory Control”, CL Programs and Multivariable Constraint Control were integrated into the block called “Process Unit Constraint Controller” and the overall system control is now coordinated through the block called “Production Rate Controller” supervisory system.

The new controls are being applied to Train #2 and the dashed blue line details the area currently

The objective of the control strategy is to drive the production rate to the maximum available value while minimizing the effects of ore starvation and ore grade problems. The PRC predictive model of the mine obtains data from the mine dispatch system to match the extraction production rate with the mine ore delivery rate. This maximizes truck utilization, maximizes extraction production rate and minimizes the risk of ore starvation.

Phase I addresses the ore delivery system from the mine to extraction and focuses on the apron feeder and ore delivery system.

As the project progresses into Phase II the balance of extraction process will be integrated into the production rate controller to manipulate the feed rate to the maximum rate allowed by the entire process unit.

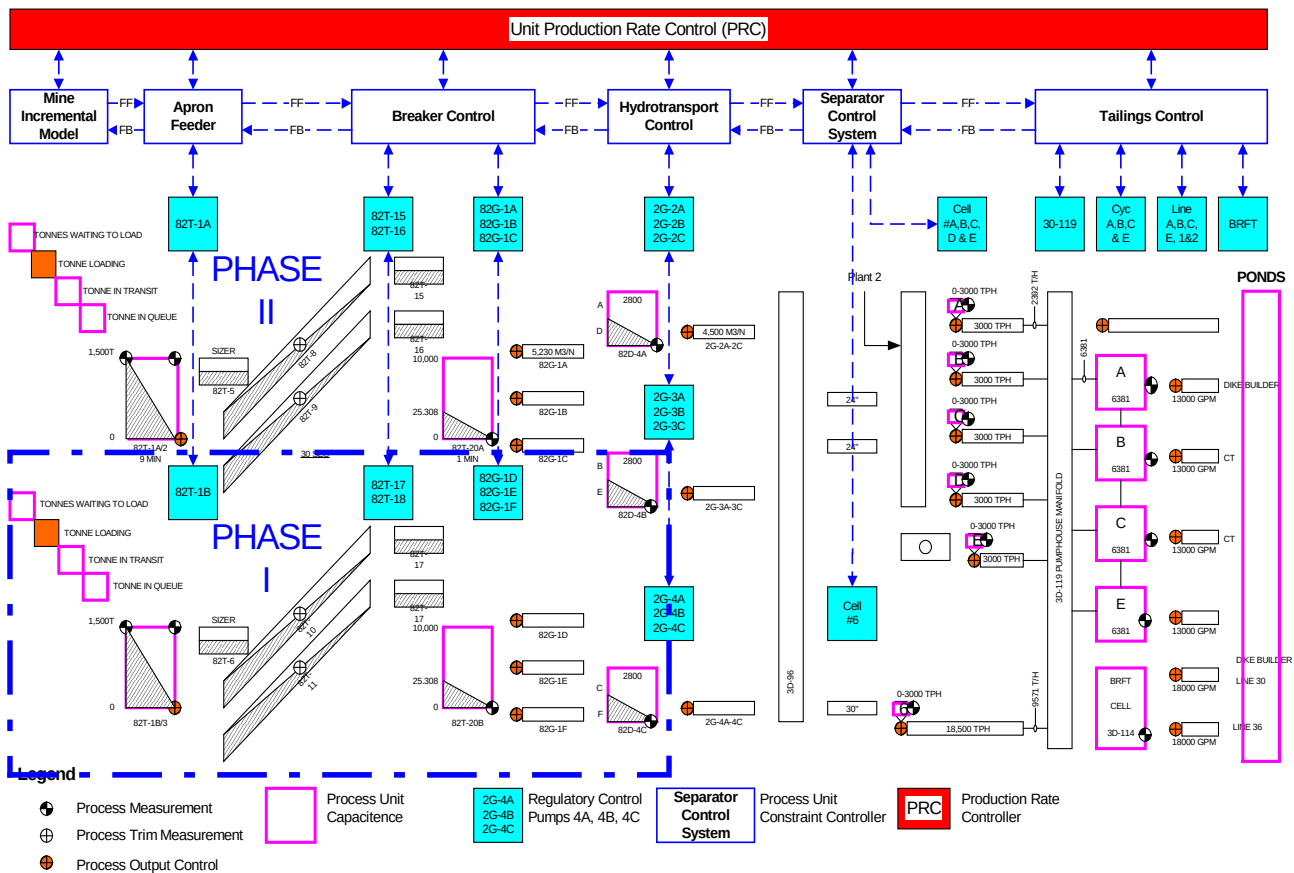


Figure 4 - Control System Algorithm

ADVANCED PROCESS MANAGEMENT TECHNIQUE

During the course of the project, another tool was required to validate the control system methods and techniques. A modeling technique of “Advanced Process Management” was implemented to capture the process and operational dynamics of the mine, extraction process and control systems. This technique was discussed in detail in a previous paper⁵ and is used to reduce the cost of the control system development and reduce the risk of errors during the control system implementation. The technique uses a simplified process model, shown in Figure 5, that allows testing control system and operating strategies off line prior to implementation on the actual process. Through the course of the project, it provided several additional benefits not seen before:

- The application of simulation packages with simplified viewers allow simulation packages to be Emailed between sites for discussions with

people outside the project team.

- The simulation became an “information vessel” that holds the knowledge of the project team members.
- There are multiple simulation packages in the market but the most effective ones allow the user to integrate into other office suites. Of special significance was the ability to capture process data online and use that to test operating scenarios.
- The ability of faster than real time dynamic testing allowed the user to simulate months of mine activity in a very short time. The control strategy developed below incorporates a state space working model of the mine shovel and truck dynamics, complete with break time and shift change emulation.
- Once the information was captured and validated, it will become a conduit to generate project training simulators and production scheduling tools at a reduced cost.

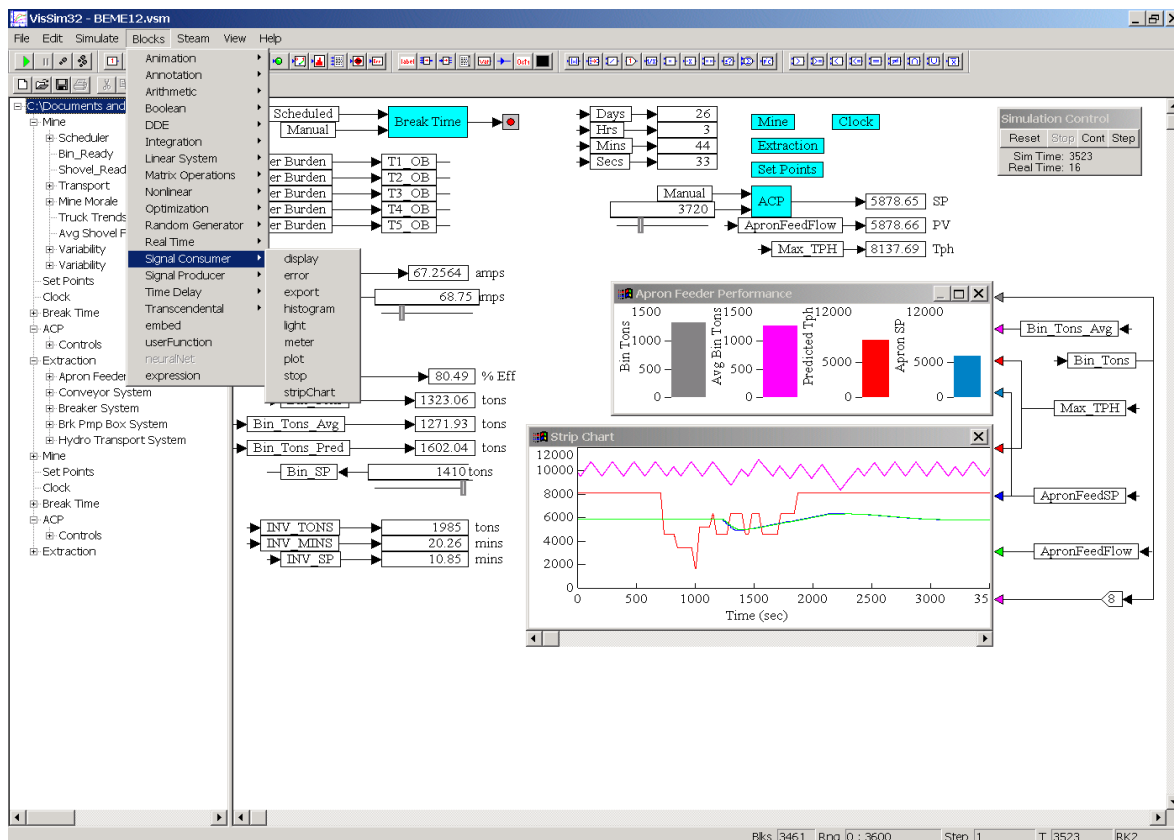


Figure 5 - Advanced Process Management Model

RESULTS

To focus the project teams, a very simple measurement target was developed. The target was to increase tonnage through the hydrotransport system and improve density control to increase the average operating density of the slurry to 1.55 SG under normal operating conditions. The anticipated outcome is the increase in production and the secondary advantages of stabilizing the slurry flow rates to decrease the hot water energy used per metric tonne of ore transported. Figure 6 is the 2000 and 2001 composite tonnage and density of the hydrotransport system that consists of three 3.8 km transport lines for moving ore from the mine to the extraction plant. A concentrated effort by

several teams began in 2001 to increase the tonnage, increase the average density of the hydrotransport slurry and decrease hot water usage. The results are summarized in the tables included with the diagrams. In 2000 the composite density was 1.49 at 46.1MM Metric Tonnes transported. During 2001 several initiatives were started resulting in the composite average density of 1.53 at 56.5MM Metric Tonnes.

The cost savings in hot water cost is calculated in the following equations. Equation 1 through 3 develops the slurry components, Equation 4 and 5 the water usage, Equation 6 the water usage reduction and Equation 7 the annual savings of C\$8.7MM.

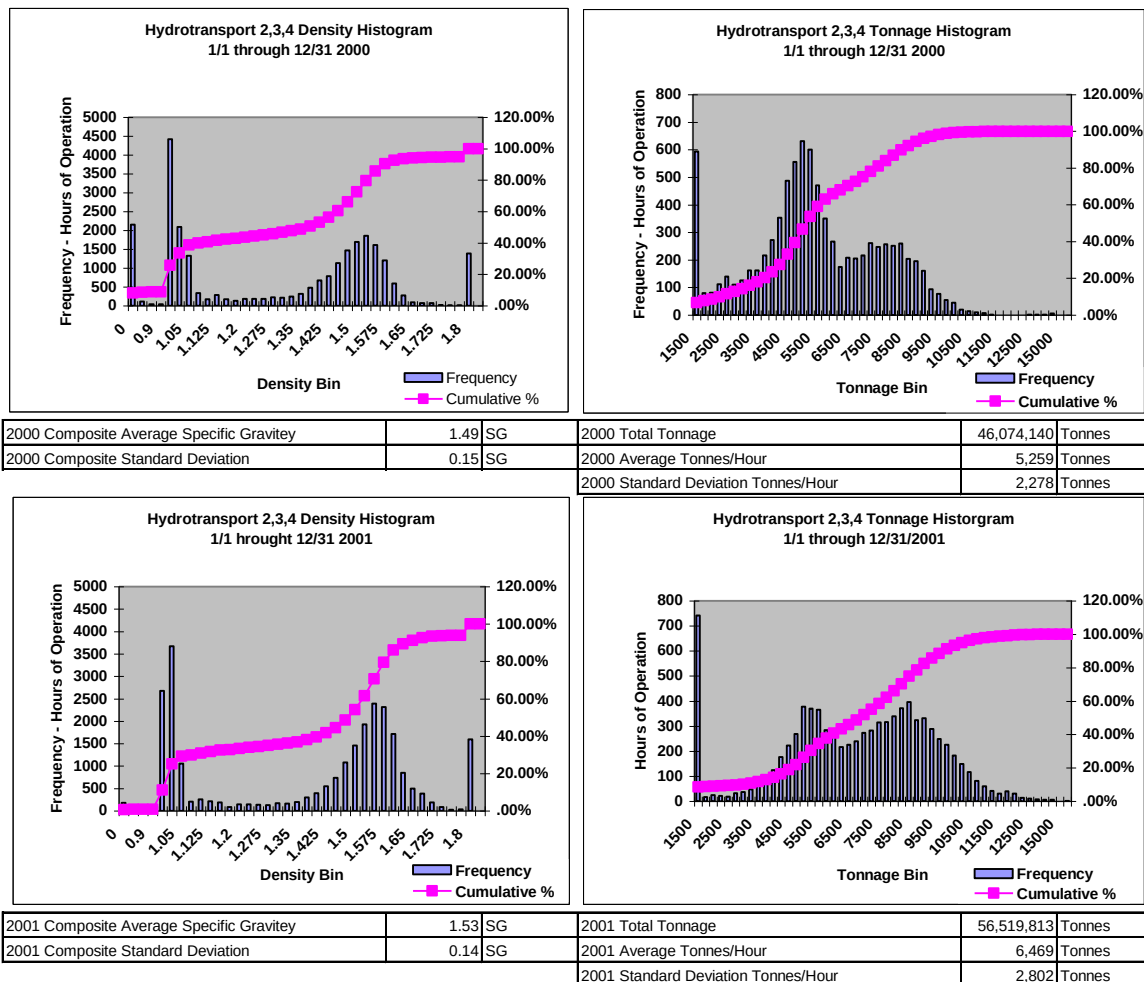


Figure 6 - Hydrotransport Density and Production Data

Given:

Tar Sand Composition:

11.50% bitumen content by weight	0.97 Specific Gravity at 200 Degrees F
4.50% water content by weight	0.97 Specific Gravity at 200 Degree F
84.00% mineral content by weight	2.65

Resulting Hydrotransport System Rates:

2000 Composite Specific Gravity at 1.49 SG at 46.1MM dry metric tonnes ore.

2001 Composite Specific Gravity at 1.53 SG at 56.5 MM dry metric tonnes ore.

Hot Water Cost Basis:

C\$8.02 per 1000 gallons 200 Degree F

Ore component Specific Gravity Calculation:

$$\left(\frac{100}{\left(\frac{84\%}{2.65SG} \right) + \left(\frac{11.5\%}{0.97} \right) + \left(\frac{4.5\%}{0.97} \right)} \right) = 2.07SG \quad \text{Equation (1)}$$

Specific Gravity Calculations – 2000 Composite Gravity Calculation 1.49SG results in a slurry component makeup by weight - %Water = 34.3% and %Ore = 65.7%:

$$\left(\frac{100}{\left(\frac{\%Water}{0.97} \right) + \left(\frac{100 - \%Water}{2.07} \right)} \right) = 1.49SG \quad \text{yields} \quad 34.3\% \text{ Water} \quad \text{Equation (2)}$$

Specific Gravity Calculations – 2001 Composite Gravity Calculation 1.53SG results in a slurry component make up by weight - %Water = 31.1% and %Ore = 68.9%:

$$\left(\frac{100}{\left(\frac{\%Water}{0.97} \right) + \left(\frac{100 - \%Water}{2.07} \right)} \right) = 1.53SG \quad \text{yields} \quad 31.1\% \text{ Water} \quad \text{Equation (3)}$$

Water Usage Calculations – 2000 with Composite Density 1.49 for 1 Metric Tonne (1000KG) of ore

$$\left(\frac{1000Kg}{1TonneOre} \right) \left(\frac{100\%Slurry}{65.7\%Ore} \right) \left(\frac{34.3\%Water}{100\%Slurry} \right) \left(\frac{1.03Meter^3}{1000Kg} \right) \left(\frac{1USGallon}{.0038Meter^3} \right) = 142Gallons \quad (4)$$

Water Usage Calculations – 2001 with Composite Density 1.53 for 1 Metric Tonne (1000KG) of ore

$$\left(\frac{1000Kg}{1TonneOre} \right) \left(\frac{100\%Slurry}{68.9\%Ore} \right) \left(\frac{31.1\%Water}{100\%Slurry} \right) \left(\frac{1.03Meter^3}{1000Kg} \right) \left(\frac{1USGallon}{.0038Meter^3} \right) = 123Gallons \quad (5)$$

Hot Water Cost Savings as a function of composite density between 2000 and 2001:

$$(123Gallons / Tonne - 142Gallons / Tonne) \times \left(\frac{C\$8.02}{1000Gallons} \right) = -C$.15 / Tonne \quad \text{Equation (6)}$$

Total Hot Water Savings for 2001 Production Run:

$$-C$.15 \times 56.5MMTonnes = C\$8.7MM \quad \text{Equation (7)}$$

CONCLUSION:

This project has just begun but three items have become apparent:

1. Suncor Energy is a unique company. The business organization has developed mechanisms to provide both the shareholders and employees rewards through programs tied to the increasing profitability of the company. The result of projects like this one is to provide a profit enhancement to the business unit that rolls to the bottom line. The team members are very much aware of the long term implications of the project success.
2. The project seeks opportunities of “Infinite Return”. This is the

application of low cost methods and technologies such as software applications and improved sensors that result in large benefits at minimal cost.

3. The application of this approach allows multiple benefits to other organizations. An example of this is the documentation effort has reduced the effort of finding and communicating control information time by 25%. This information was integrated into the training packages to enhance the training documentation. That training documentation has now become a major resource to the control project.

This very exciting project has just started, and there is no limit to the opportunities ahead.

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⁵ Robinson, James; *Advanced Process Management and Implementation*, IETC 1999

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